

US Baby Names

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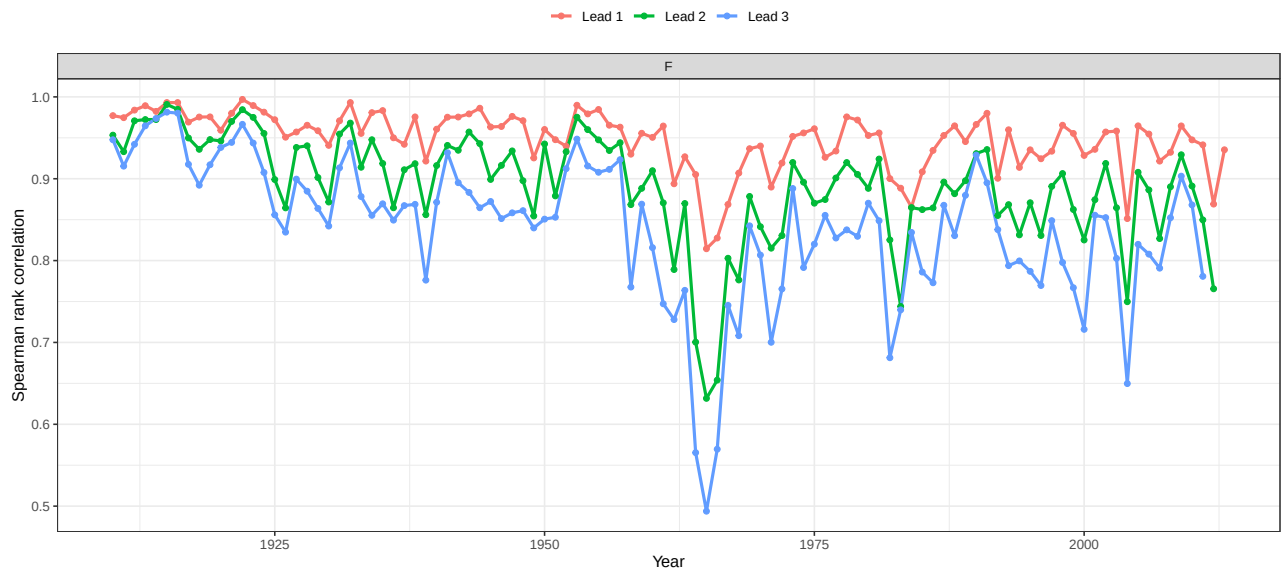
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0.1. Spearman Correlation by Year

0.1.1. Spearman Correlation for Female Names

Persistence of Popular Names: F

Spearman rank correlation between each year's top 25 names and the next 1–3 years

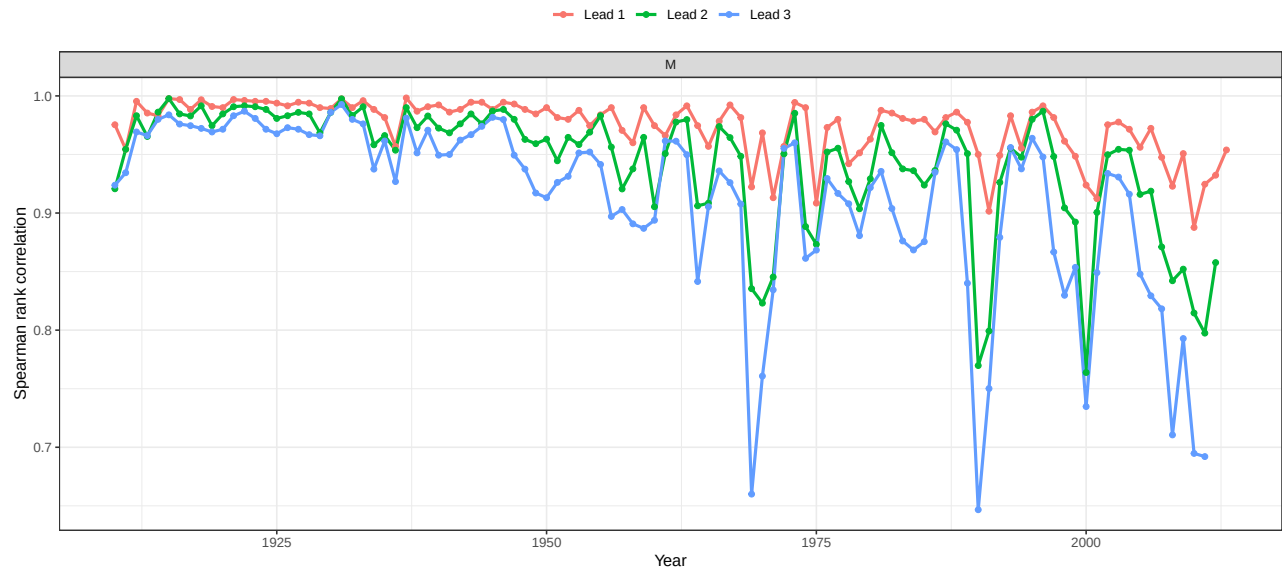


The graph shows that the top 25 female names are very persistent for the time period analysed. For most of the years, the Spearman rank correlation is between 0.8 and 1, except for between 1960 and 1970, around 1985 and around 2005. Therefore, if a female name is in the top 25, then there is a very high probability that it will be in the top 25 in the following year as well as the year after that. According to this graph, it appears that female names are no more or less likely to persist since the 1990s compared to earlier years; the pattern does not appear to change significantly from the 1990s.

0.1.2. Spearman Correlation for Male Names

Persistence of Popular Names: M

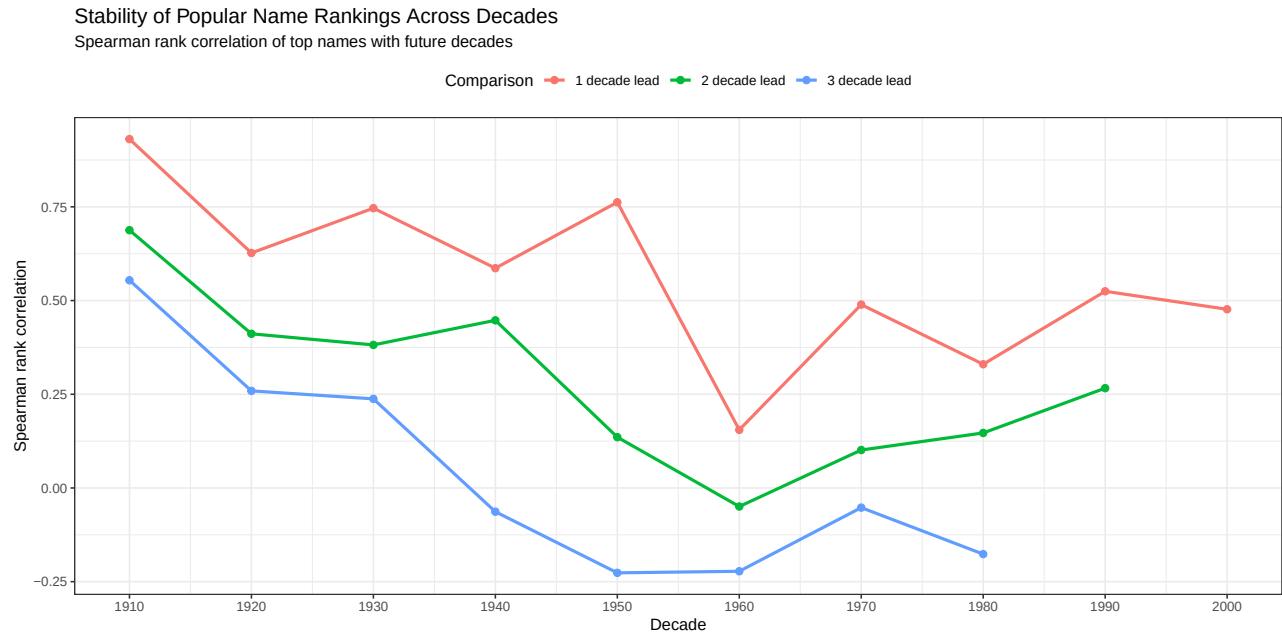
Spearman rank correlation between each year's top 25 names and the next 1–3 years



The graph shows that the top 25 male names are very persistent for the time period analysed. Before 1965, the Spearman rank correlation remains between 0.85 and 1, however, the range significantly increases and the consistency decreases after 1965. This indicates that male names were initially quite persistent, but that persistence has decreased significantly since 1965. Therefore, the probability of a male name remaining in the top 25 male names began very high, but in more recent years, that probability has become smaller. From this graph, it appears that male names are less persistent since the 1990s than prior to that - popular names have become a bit slower to persist since the 1990s.

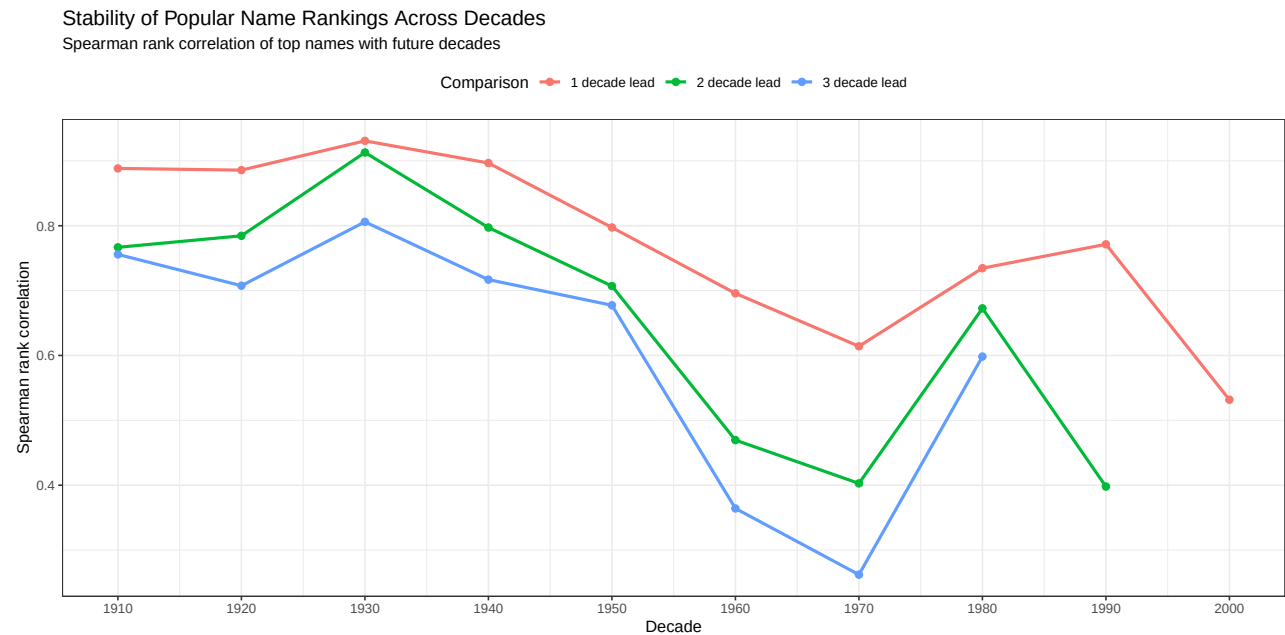
0.2. Spearman Correlation by Decade

0.2.1. Spearman Correlation by Decade for Female Names



The same trends are apparent when looking at popular female names by decade - according to this graph, it appears that female names are no more or less likely to persist since the 1990s compared to earlier years; the pattern does not appear to change significantly from the 1990s.

0.2.2. Spearman Correlation by Decade for Male Names



Once again, this graph shows similar trends to the yearly popular male names graph above. From this graph, it appears that male names are less persistent since the 1990s than prior to that - popular names have become a bit slower to persist since the 1990s.

0.3. Names Surge

The table below shows the top 10 female names and top 10 male names that experienced the largest year-to-year percentage rank improvement. The table excludes names with an initial count of less than 100, so that the rank improvement is not driven by relative large changes in very unpopular names.

Table 0.1: Top 10 year-on-year name surges by gender

Year	Gender	Name	Previous count	Current count	Change in count	% change in count	Previous rank	Current rank	Rank improvement
1976	F	Jaime	897	7836	6939	773.6	262	29	803.4483
1989	F	Kiara	171	2595	2424	1417.5	832	115	623.4783
2002	F	Ashanti	253	2927	2674	1056.9	784	115	581.7391
1957	F	Tammy	204	4363	4159	2038.7	602	107	462.6168
1979	F	Brianne	133	1655	1522	1144.4	813	168	383.9286
1995	F	Alondra	111	1149	1038	935.1	1121	242	363.2231

Year	Gender	Name	Previous count	Current count	Change in count	% change in count	Previous rank	Current rank	Rank improvement
2007	F	Miley	141	1213	1072	760.3	1249	277	350.9025
1966	F	Michelle	16215	27152	10937	67.4	18	4	350.0000
1991	F	Iesha	253	1875	1622	641.1	703	157	347.7707
1982	F	Kayla	266	2268	2002	752.6	583	132	341.6667
2011	M	Mason	14831	19488	4657	31.4	12	2	500.0000
2013	M	Jayceon	128	1827	1699	1327.3	1059	208	409.1346
2010	M	Bentley	497	3762	3265	656.9	505	100	405.0000
1998	M	Dawson	141	1883	1742	1235.5	779	175	345.1429
2013	M	Noah	17302	18179	877	5.1	4	1	300.0000
1995	M	Tristan	441	3085	2644	599.5	454	121	275.2066
2013	M	Jase	1100	4539	3439	312.6	304	89	241.5730
2002	M	Ethan	17959	22105	4146	23.1	17	5	240.0000
1995	M	Bailey	112	990	878	783.9	853	273	212.4542
1972	M	Christopher	48251	52195	3944	8.2	6	2	200.0000

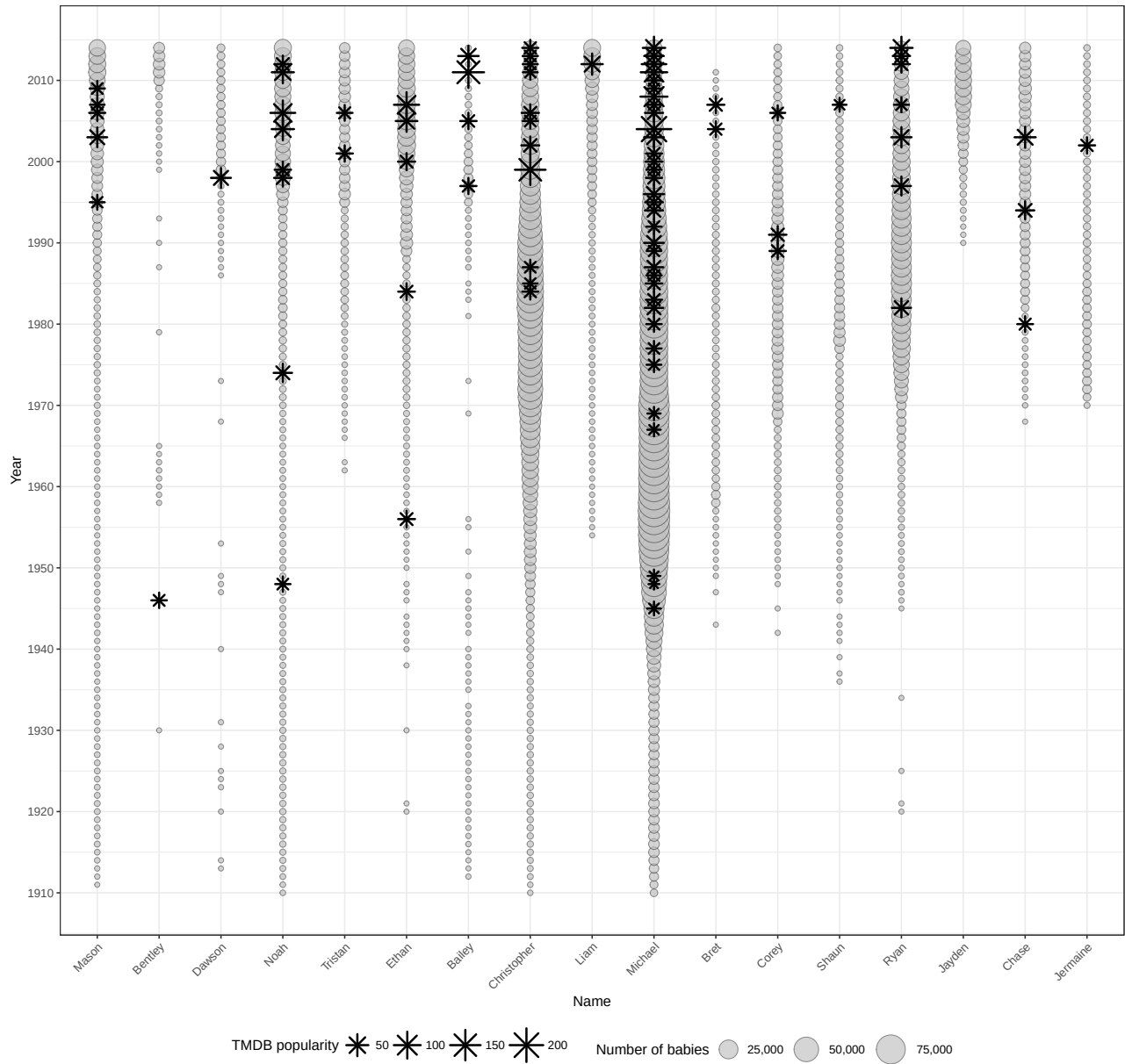
0.4. Names Surge and TV Shows/ Movies

0.4.1. Male Names

The plot below shows how the popularity and persistence of male names interacts with character names of TV show or movie characters. The male names chosen are the 20 male names that experienced the largest percentage rank improvement. These names were then merged with character names.

Baby-name popularity and media releases: M

Top 25 matched names. Bubble size = number of babies; star size = TMDB popularity. Years end at 2014.



As seen from the plot above, there is not a clear pattern between the popularity or persistence of names and character names from TV shows/ movies for these particular male names. However, it is interesting to observe how these variables develop and interact over time. Also, for most of these names, their popularity has been increasing over time. This makes them an excellent choice for a toy company, because their popularity is increasing, and increasing regardless of popular character names. It is clear that some names are classics - Christopher, Michael and Ryan remain persistently popular, regardless of whether TV show or movie characters have these names. These names are also the names of characters in many TV shows/ movies. Therefore, for a toy company, these names are good to take

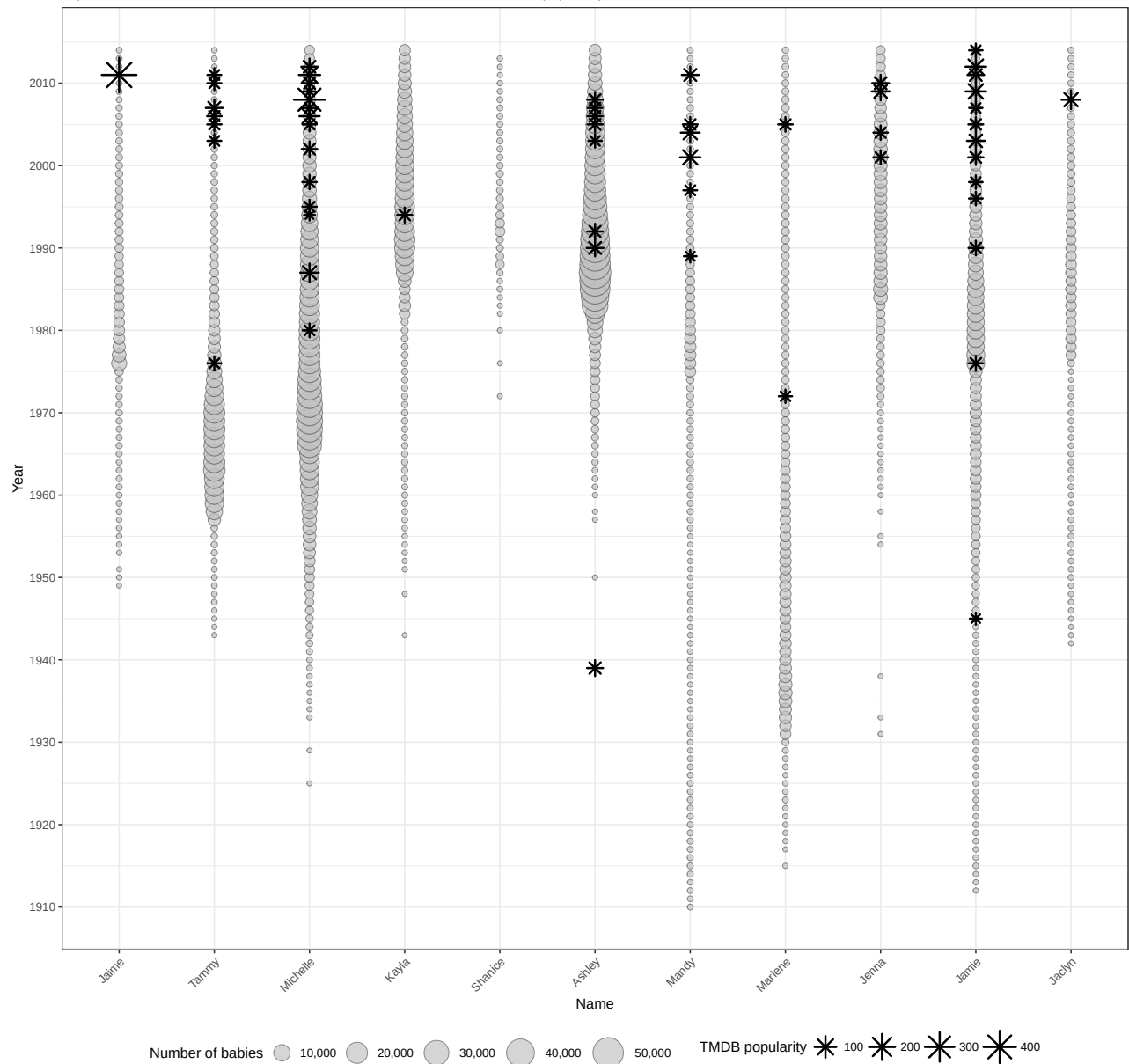
note of, because they are unlikely to change significantly in popularity over time, meaning they are a lower-risk investment.

0.4.2. Female Names

The plot below shows how the popularity and persistence of female names interacts with character names of TV show or movie characters. The female names chosen are the 20 female names that experienced the largest percentage rank improvement. These names were then merged with character names.

Baby-name popularity and media releases: F

Top 25 matched names. Bubble size = number of babies; star size = TMDB popularity. Years end at 2014.



As seen from the plot above, there is not a clear pattern between the popularity or persistence of names and character names from TV shows/ movies for these particular female names. However, it is interesting to observe how these variables develop and interact over time. It is clear that some names are classics - Michelle, Kayla, Ashley, Jenna and Jamie remain persistently popular, regardless of whether TV show or movie characters have these names. Therefore, for a toy company, these names are good to take note of, because they are unlikely to change significantly in popularity over time, meaning they are a lower-risk investment.